AD 502 – Machine Learning

Pre-Reading Material – Lecture 30

Neural Network-I

Introduction to Neural Networks

A neural network is a machine learning model inspired by the functioning of the human brain. It consists of interconnected processing units called artificial neurons that learn patterns from data and generate predictions or outputs.

Neural networks are an important part of artificial intelligence and are used in image recognition, speech processing, forecasting and classification.

What Is an Artificial Neuron?

An artificial neuron is the basic processing unit of a neural network. It receives input information, processes it and produces an output.

The main components of an artificial neuron are:

- Inputs: Data provided to the neuron.
- Weights: Values representing the importance of different inputs.
- Bias: An adjustable value that influences the neuron's output.
- Activation function: Determines the neuron's final output.
- Output: The result produced by the neuron.

Working of a Neural Network

A neuron receives input data and combines the inputs according to their weights and bias. An activation function processes the resulting value to generate an output. During training, the network adjusts its parameters to improve its predictions.

For example, a neural network can predict student performance using information such as attendance, assignment scores and previous examination marks.

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Applications of Neural Networks

Neural networks are widely used in:

- Image and facial recognition.
- Speech recognition.
- Medical diagnosis support.
- Weather and electricity-demand forecasting.
- Text classification and language processing.

Advantages and Limitations

Neural networks can learn complex patterns and model nonlinear relationships. However, they may require large amounts of data, considerable computational resources and careful training.

Conclusion: Neural networks are fundamental machine learning models that learn patterns from data through interconnected artificial neurons.

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